Substance Abuse Detection AI

Data Intelligence + Decision Support

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PROBLEM

A Crisis Without Real-Time Surveillance

The U.S. substance abuse crisis causes **80,000+ deaths annually**, yet traditional surveillance — hospital records, population surveys — carries lag times of **months to years**.

Online forums hold real-time, high-volume discourse on substance use, distress, and addiction. But this text is noisy, slang-heavy, and entangled with personally identifiable information.

We bridge this gap with a **privacy-preserving, population-level decision-support system**: ingest social text → detect risk signals → correlate with CDC data → surface analyst-ready summaries.

DATA

Five Public Datasets

DATASET	SIZE / SCOPE	PIPELINE ROLE
UCI Drug Reviews Kaggle CSV	41,934 posts	Training · risk labels · embedding baseline
Erowid Experience Vault Public scrape	~5,000 posts	Slang lexicon · LSA topic anchors
CDC Overdose API data.cdc.gov	2015–2024 monthly	Ground-truth for lead-lag correlation
NSDUH SAMHSA	Annual national	Prevalence normalization
NIDA Summary Tables Federal data	Annual per-100k	Macro-baseline for dashboard reporting

PREPROCESSING

Four-Stage Privacy Pipeline

Raw text passes through four stages before any model sees it:

. **HTML & Noise Removal** — BeautifulSoup + regex strip markup, URLs, special characters

. **PII Scrubbing** — spaCy NER masks names, locations, phones, e-mails; no identity stored

. **Slang Normalization** — Curated lexicon maps colloquialisms to standard substance tokens

. **Deduplication** — Semantic hashing removes near-duplicates (Jaccard ≥ 0.85)

ETHICS

Privacy-Preserving by Design

Anonymization
Irreversible PII masking at ingestion — no identity stored or transmitted

Population-level only
Dashboard architecturally blocked from surfacing individual records

Non-punitive design
Scoped to public health planning; structurally incompatible with law enforcement

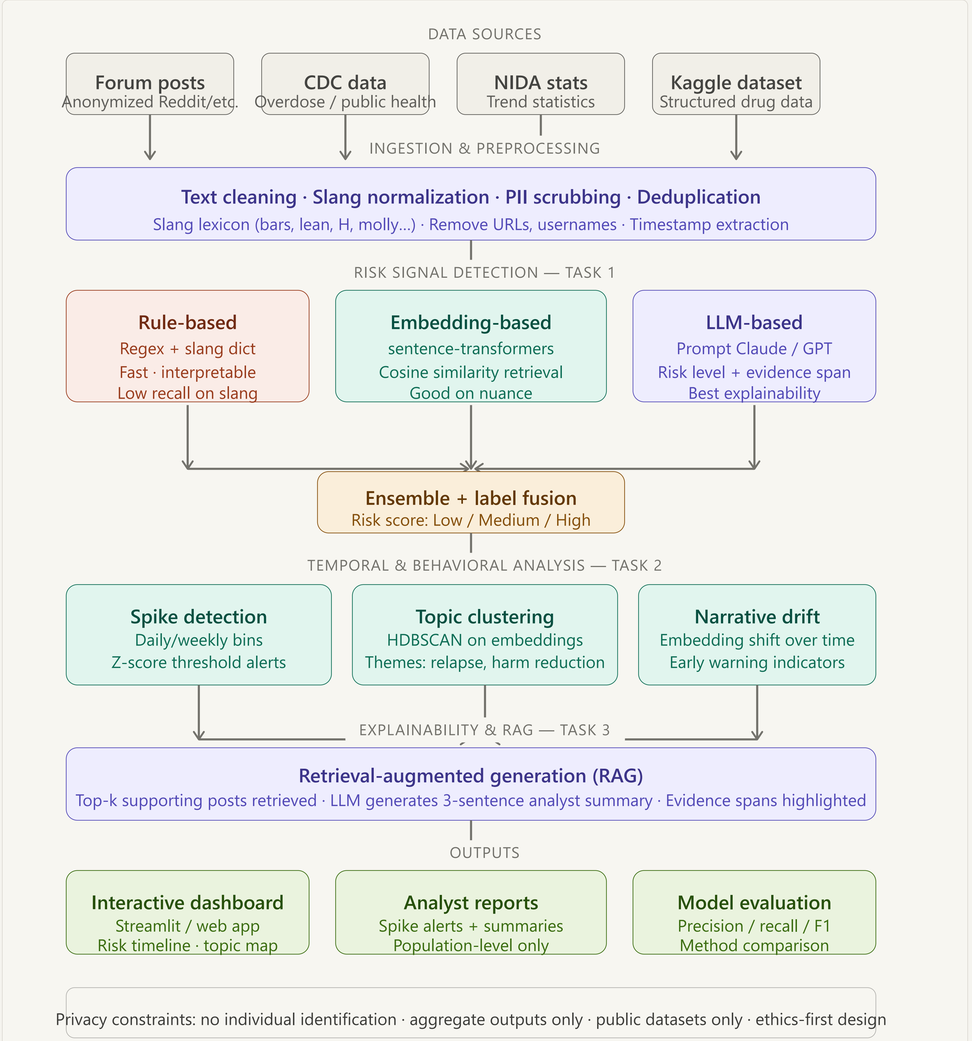
Human-in-the-Loop
All signals labeled for analyst review — final decisions remain with humans

Bias-aware
Lexicon and seed-bank biases documented; HITL audit flags systematic errors

Transparent
All code, datasets, and model decisions open-source and auditable

SYSTEM ARCHITECTURE

End-to-End Pipeline



METHODS

Four-Method Ensemble

METHOD	TECHNOLOGY	WEIGHT	KEY STRENGTH
Rule-Based	Regex + slang lexicon	0.25	Deterministic; high precision on known slang
Embedding	Sentence-BERT + FAISS	0.30	Cosine similarity to labeled seed bank
LLM Classifier	Gemini Flash	0.35	Zero-shot + evidence span extraction
Fine-Tuned BERT	DistilBERT (41K posts)	0.10	Captures nuanced contextual dependencies
RAG Explainability	FAISS + Gemini Flash	—	Top-k evidence retrieval → analyst summaries

K-Means (k=8) clustering on UMAP-reduced embeddings identifies evolving behavioral sub-populations. Temporal spike detection uses rolling z-scores (threshold Z > 2.5) binned monthly.

KEY FINDING

Early Warning Signal

−3 mo

Opioid social signals lead CDC overdose counts by **up to 3 months** — a statistically significant early-warning indicator for public health response

Pearson r **0.471** p-value **0.006** MRR (opioid) **0.298**

CLASSIFICATION RESULTS

Held-out Test Set (n = 2,000)

METHOD	ACCURACY	MACRO-F1	ROC-AUC
Rule-Based	0.495	0.303	0.504
Embedding (SBERT)	0.325	0.295	0.490
Fine-Tuned BERT	0.404	0.280	0.481
Ensemble	0.404	0.304	0.506

Test set: low-risk 1,014 / medium 791 / high 195 (9.75% high-risk). Benzo detection strongest (ensemble F1 = 0.366). High-risk recall is hardest due to class imbalance.

CLUSTERING

Behavioral Sub-populations (k = 8)

0.960
Cluster-level NDCG — excellent risk ranking across 8 thematic groups

60%
HITL approval rate — 6 of 10 audited analyst outputs approved by domain reviewers

Eight clusters span distinct substance families (cannabis, xanax/benzos, opioids, stimulants) and behavioral contexts (harm-seeking vs. harm-reduction). Silhouette = 0.082, consistent with overlapping drug discourse.

EXAMPLE SYSTEM OUTPUT

RAG Analyst Summary

> *A 300% surge in high-risk opioid signals was detected across Cluster 0 (4,617 posts) between October 12–19. Discourse features fentanyl analogs mislabeled as prescription stimulants. Recommended action: Alert harm-reduction networks to distribute updated test strips and issue clinician advisories.*

All summaries are aggregate and population-level. Individual posts are never surfaced. Final intervention decisions remain with healthcare professionals.

FUTURE WORK

Roadmap

- **Live stream integration** — Reddit/Twitter APIs for near-real-time monitoring rather than batch processing
- **Dynamic lexicon updates** — Auto-surface emerging slang (e.g. xylazine-class terms) from high-risk clusters
- **Geospatial heatmaps** — County/state-level risk maps from anonymized location tokens
- **Active HITL feedback loop** — Analyst approve/reject votes feed back into BERT fine-tuning
- **Multimodal expansion** — Image signals from pill photography and harm-reduction meme culture